Project Check-in 1: Sketch-to-Photo Synthesis with ControlNet-Guided Diffusion

CMPE 261: Generative AI

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Finalized Specifications

We are keeping the scope from our proposal. We will use Stable Diffusion 1.5 with pretrained ControlNet (canny and scribble) checkpoints through Hugging Face Diffusers as the baseline, then fine-tune ControlNet with LoRA on a small set of real hand-drawn sketches. We will compare the baseline against the fine-tuned model on the same sketches and run ablations on the ControlNet conditioning scale and the guidance scale. Results will be measured with FID against real COCO images, CLIP score for prompt alignment, and a small pairwise human preference survey. Training and batch inference will run on the COE HPC.

Tentative Abstract

Turning a rough sketch into a realistic image is useful for fast prototyping and concept design. Pretrained ControlNet models work well on clean edge maps, but they tend to struggle with real freehand sketches, which have broken lines, uneven strokes, and missing detail. In this project we study how well a Stable Diffusion plus ControlNet pipeline transfers from synthetic edge maps to real hand-drawn sketches. We build a dataset of COCO images paired with Canny and HED edge maps, plus a small set of hand-drawn sketches, and fine-tune ControlNet with LoRA on the hand-drawn set. We then compare the baseline and fine-tuned models with FID, CLIP score, and a human preference survey, and look at how conditioning and guidance scale affect the output. We expect fine-tuning to improve how closely the image follows the sketch without hurting realism, and we will report where it still fails.

Data Sources and Initial Exploration

Our photos and captions come from a subset of COCO 2017 (about 2,000 to 5,000 images), and the captions double as text prompts. We generate synthetic sketches from these images using Canny and HED edge detection, which follows the same pairing idea ControlNet was trained with. We are also drawing 50 to 100 sketches by hand, each with a short prompt, to use for fine-tuning and testing. So far, we have been checking category balance and image sizes in the subset, resizing to 512 x 512, trying different Canny thresholds and HED to see which edges look closest to real sketches, and running the pretrained baseline on a few inputs to see where it breaks. Initial exploration showed a trade-off when selecting the Canny threshold: lower thresholds preserved more thin and broken contours but introduced additional noise, while higher thresholds produced cleaner edge maps but removed some weak details. HED generally produced more continuous object contours than Canny, although it sometimes retained background or texture boundaries. In our initial tests, the pretrained ControlNet followed clean synthetic edge maps more reliably than incomplete hand-drawn sketches, motivating our planned LoRA fine-tuning using real sketches.

Project Management

Our code lives in a GitHub repo at https://github.com/LRozay/Sketch-to-Photo and the instructor (bertin.cordovadiba@sjsu.edu) and TA (charansai.gandham@sjsu.edu) have been invited. The repo is organized into data scripts, training and inference code, exploration notebooks, evaluation code, and the report.

Report Outline

The report will have these sections: abstract, introduction and motivation, related work (latent diffusion, ControlNet, LoRA), data, method, experiments and evaluation setup, results, discussion and failure cases, conclusion and future work, and references.

Next Steps

Before Check-in 2 we will finish the hand-drawn and synthetic datasets, get the baseline running end to end on HPC and record baseline FID and CLIP scores, start the first LoRA fine-tuning runs, and draft the introduction, related work, and data sections of the report.